

Pavle Josić

Full-Stack Developer • Belgrade, Serbia

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PROFILE SUMMARY

Software and Information Engineering student with a strong specialization in backend engineering, distributed systems, and application security (AppSec). Stack-agnostic developer experienced in building scalable server-side architectures, implementing defensive design patterns, and bridging the gap to frontend operations using modern frameworks to ensure seamless end-to-end performance.

TECHNICAL ARSENAL

LANGUAGES: JavaScript • TypeScript • Python • Solidity • Java

FRAMEWORKS, LIBS & TOOLS: Next.js • React • Nest.js • Spring Boot • Express.js • Tailwind CSS • GSAP • Claude AI • Postman

BACKEND, OPS & DBS: Node.js • PostgreSQL • Prisma ORM • Docker • Redis

SELECTED WORKS

Microservices Restaurant Management System

[GitHub](#) →

Spring Boot • Java • PostgreSQL • Docker • Microservices • JWT

Architected and deployed a distributed backend ecosystem split into decentralized microservices for decoupled authentication and core enterprise operations. Utilized containerized PostgreSQL instances to enforce strict database-per-service isolation, integrated a centralized JWT validation layer, and established strict Role-Based Access Control (RBAC) to harden inter-service communication mesh.

Decentralized Automated Market Maker (AMM)

[GitHub](#) →

Solidity • Smart Contracts • ERC20 • OpenZeppelin • DeFi Architecture

Engineered a Uniswap V2-style decentralized financial exchange deployed on Ethereum testnets. Implemented the Factory-Pair-Router structural pattern for deterministic and dynamic liquidity pool deployments. Developed smart contracts managing LP token minting through the Constant Product Formula and governed reliable on-chain token swaps backed by automated protocol fee generation layers.

Tehnički Pregled Lazarević 1968

[Live Link](#) →

Next.js • GSAP • Tailwind CSS • Vercel

Developed and shipped a live web platform for a technical inspection business with a strict focus on high-performance UX. Engineered responsive, fluid layout sequences using GSAP and Tailwind CSS. Implemented advanced Search Engine Optimization (SEO) strategies to maximize local discovery, search crawl indexing performance, and keyword ranking metrics.

NNCraft: Deep Learning Framework from Scratch

[GitHub](#) →

Python • NumPy • Linear Algebra • Machine Learning

Built a dense neural network framework from scratch using pure NumPy to isolate and analyze foundational mathematical mechanics of machine learning. Authored abstractions for custom network layers, designed manual matrix-based backpropagation pipelines, and natively implemented optimized training constraints including Adam, RMSprop, and regularizers.

EDUCATION & TIMELINE

Software and Information Engineering

Singidunum University (3rd year)

2023 – Present

Information Technology

Electrotechnical School "Rade Končar"

2019 – 2023